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液冷电源托盘与 800 VDC

Jared Huntington（NVIDIA）
Roya Movahedi · 石磊（MEGMEET）

时长 18 分 43 秒

来源 youtube.com/watch?v=UONmRpabnf4

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Jared Huntington（NVIDIA）、Roya Movahedi / 石磊（MEGMEET）OCP Global Summit 2026 演讲 · 图文实录

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- **图片**：演讲当时投影的幻灯片，按时间顺序
- **中文译文**：讲者在这一页停留期间所说的话，逐句翻译，不作删减
- **英文原句**：同段话的英文原文，供核对与引用
- **小结**：整理者对这一段的一句话归纳，与讲话内容明确区分

原文由 YouTube 自动字幕转写，已去除 um、uh 等口头填充词，并修正明显的转写错误。本片字幕对专有名词与单位的识别错误较多，统一修正如下：**MEGMEET**（字幕作 MagMates / MagNet's）、**800 V / 54 V**（字幕多作 800 W / 54 watt）、**MGX**（字幕作 MTX）、**SST 固态变压器**（字幕作 SSD）、**2U**（字幕作 2 RU）、**Shi Lei 石磊**（字幕作 Shilei）。

凡口播数字与幻灯片标注不一致处，正文采用幻灯片值，并在当节注明差异；仅存在于口播、幻灯片上无对应的数字，标注为「口头表述」。

中文译文以"准确传达原意"为优先，而非追求文采。

视频信息

项目	内容
标题	Liquid-Cooled Power Shelf and 800 VDC Platform for Next-Generation Datacenters
讲者	Jared Huntington，NVIDIA 资深电源架构师 Roya Movahedi，MEGMEET 首席营销官 石磊（Shi Lei），MEGMEET 首席工程师
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项目	内容
频道	Open Compute Project（YouTube）
链接	https://www.youtube.com/watch?v=UONmRpabnf4

第一部分　为什么必须走到 800 VDC

01　三位讲者与议题



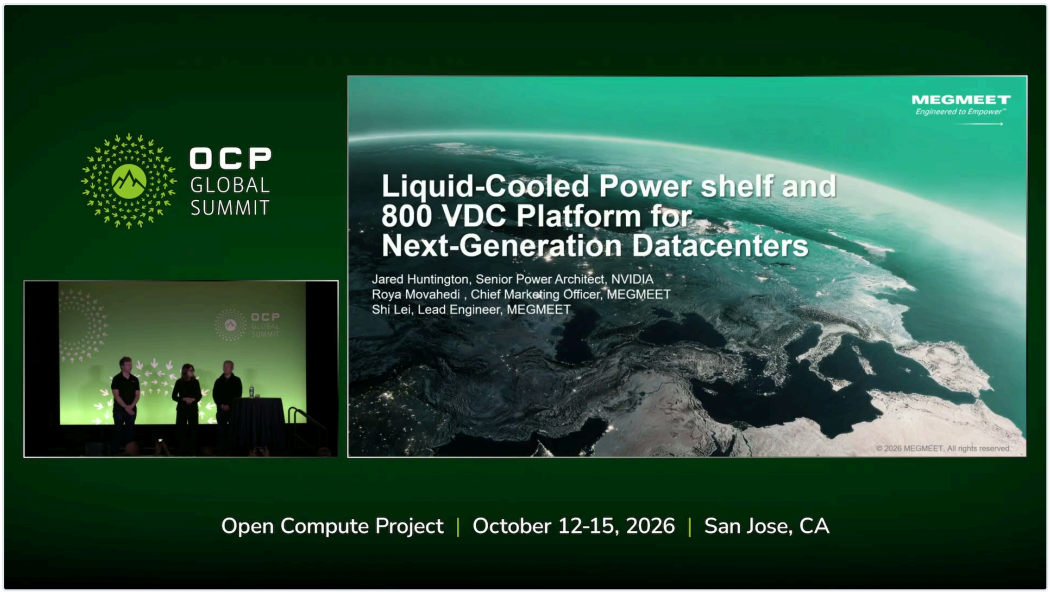
OCP Global Summit 2026 主题页：Scaling Innovation for the AI Era，October 12-15, 2026，San Jose, California

好，大家好。我是 Jared Huntington，NVIDIA 的电源架构师，做的是从 SST（固态变压器）一直到芯片这一路的电源。

> Okay. Hello everyone. So I'm Jared Huntington. I'm a power architect with Nvidia working on power basically from SST to chip and

> **小结**　"从 SST 到芯片"这句话划定了这场演讲的纵深：它不是讲一个电源模块，而是讲从变电站进线到 GPU 核心电压之间的整条链路。

02　800 VDC 的必要性



Liquid-Cooled Power shelf and 800 VDC Platform for Next-Generation Datacenters：Jared Huntington（NVIDIA）、Roya Movahedi（MEGMEET）、Shi Lei（MEGMEET）

你先来。这位是 Roya Movahedi，MEGMEET 的首席营销官。我是石磊（Shi Lei），MEGMEET 的首席工程师，主要负责液冷电源的设计。

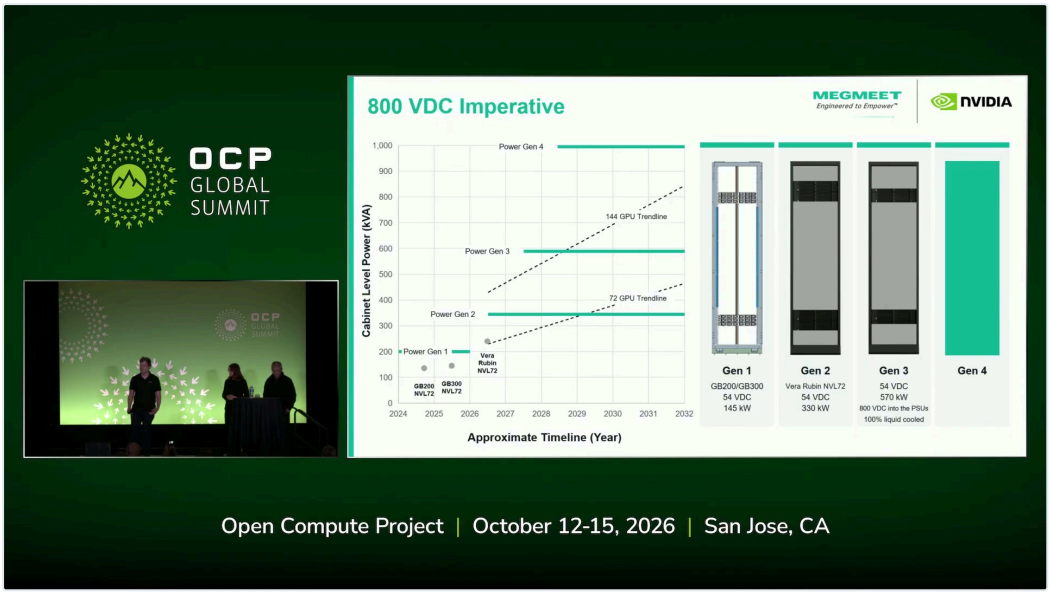
好，今天我们要讲的是液冷电源托盘，以及我们向 800 V 直流的过渡；还有 MEGMEET 与 NVIDIA 一起、为整个生态开发液冷电源托盘的工作。我们翻到下一页。

那是什么？好，那么——希望大家已经见过类似的图。我们在好几个演讲里都放过这张图，它基本上就是"为什么需要 800 V"的答案。

> I'll let you go first. Roya Movahedi, CMO at MEGMEET. Yeah, I'm Shi Lei. I'm the lead engineer in MEGMEET and focus on the liquid cool power supply design. Okay, so today we're talking about the liquid cooled power shelves and our transition to 800 V DC. And the work that MEGMEET is doing with Nvidia on developing liquid cooled power shelves that we can use for the ecosystem. So I think we go to the next. What's that? Yeah, so then hopefully everyone's seen a graph similar to this, right? We present this in several of our talks, but this is essentially what's driving the need for 800 V.

> **小结**　这是一场 NVIDIA 提需求、MEGMEET 交产品的双人演讲——电源托盘已经变成需要两家公司联合定义规格的部件。

03　四代电源平台：145 → 330 → 570 → 原生 800 V



800 VDC Imperative：机柜功率（kW）对时间（2024–2032）的四级台阶，Power Gen 1–4 与 72/144 GPU 趋势线；右侧四列机架为 Gen 1 GB200/GB300 NVL72 54 VDC 145 kW、Gen 2 Vera Rubin NVL72 54 VDC 330 kW、Gen 3 54 VDC 570 kW（800 VDC 进 PSU、100% 液冷）、Gen 4 原生 800 VDC

本质上，如果我们看功耗随时间的变化，会有逐年增长——纯粹是因为芯片、因为 GPU 为了更高性能而拉走更多电力。但 GPU 还有另一个独特之处：我们非常想把尽可能多的 GPU 塞进同一个机柜，因为只要把所有 GPU 保持在同一个铜缆网络域里，我们本质上就能提升性能。这就是 GPU 与传统 Web 服务器根本上的不同。

所以我们很快就能从 NVL72 系统，把功率翻倍到 NVL144 系统。而这种功率翻倍，只要封装一有进展，就会立刻发生。

所以作为做电源的工程师，我们不想成为系统的瓶颈。我们不想成为那个"闸门"，卡住机架里下一层性能。这就是为什么我们必须为功率水平的快速上升做好准备。

为了支撑这些计算机架，我们实际上有几代电源平台。第一代，配 GB200、GB300，能做到 145 kW。第二代基本能做到 330 kW，也就是我们在 Vera Rubin 上用的。而现在我们正在做的第三代电源平台，是 100% 液冷的电源托盘。有了它，如果一个机架完全装满、把 54 V 母排推到极限，我们可以支持到 570 kW（口播为 "50-V 母排"，幻灯片标注为 54 VDC）。

最后，我们还在做原生 800 V 的机架——把 800 V 直流直接送进机架。我们还没准备展示它长什么样，但第三代是通往那里的踏脚石。从机架电源架构的角度看，我们要做的是：把 800 V 直流送进来，然后把它……

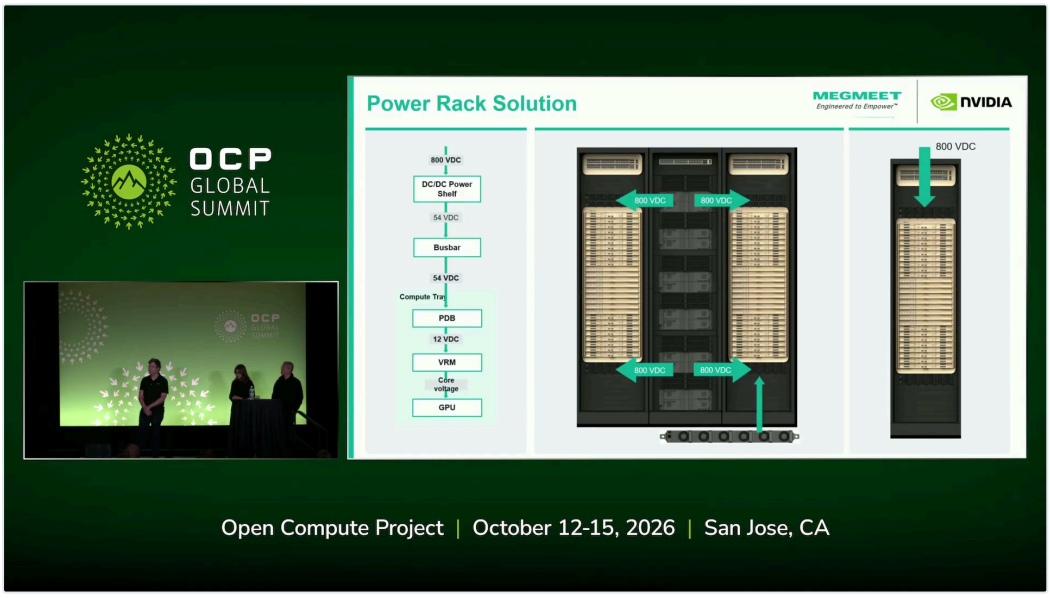
> And it's essentially if we look at the power consumption over time, right? There's a year-over-year increase just due to the chips, due to the GPUs pulling more power for more performance. But there's another aspect about GPUs that's unique, which is we really want to pack as many GPUs into one cabinet as possible because we can essentially increase the performance if we keep all of those GPUs on a copper networking domain. And so that's what's fundamentally different between GPUs and the classic web servers. So very quickly we can go from an NVL 72 system and we can double power to an NVL 144 system, right? And so that doubling of power can happen as soon as there's new developments in packaging. And so as engineers working on power, we don't want to be the bottleneck in the system. We don't want to be the gate for what's the next level of performance that we can get in the rack. And so that's where we

> need to be prepared for power levels to increase very quickly. And so in order to support these computer racks, we have several power generations. Our first generation with GB200, GB300 was capable of 145 kW. Our second generation was basically capable of 330 kW, and that's what we're using for Vera Rubin. And then now what we're working on for our generation 3 of power platforms is 100% liquid cooled power shelves. And with that, if we completely fill a rack and we push to the limits of a 54-V bus bar, we can support up to 570 kW. And then finally, we are working on native 800-V racks where we take 800-V DC directly into the rack. We're not prepared to show what that looks like quite yet, but this generation 3 is kind of a stepping stone to get us there. And so from a rack power architecture standpoint, what we're doing is we're taking in 800-V DC, and then we're converting it

> **小结**　"只要封装一有进展，功率就会立刻翻倍"——这句话把电源设计逼进了被动位置：留给供电的迭代窗口，比留给封装的短得多。

第二部分　液冷：从机架到机房

04　电源机架方案：800 → 54 → 12 → 核心电压



Power Rack Solution：左侧为供电链路 800 VDC → DC/DC Power Shelf → 54 VDC → Busbar → Compute Tray → PDB → 12 VDC → VRM → Core voltage → GPU；中、右为电源机架正反面，800 VDC 由母排分配到各电源托盘

……转换成 54 V 直流，落在 54 V 母排上。而在开发的第一个阶段，先做一个电源机架：输入交流，输出 54 V 直流。抱歉，是输出 800 V，然后我们再把 800 V 转成 54 V。

这样一来，长期看，任何需要具备 800 V 能力的东西——不管是支撑机架、网络机架，甚至我们的计算机架——只要它是围绕 MGX 母排设计的，我们都能让它落在交流数据中心，或者 800 V 直流数据中心里。

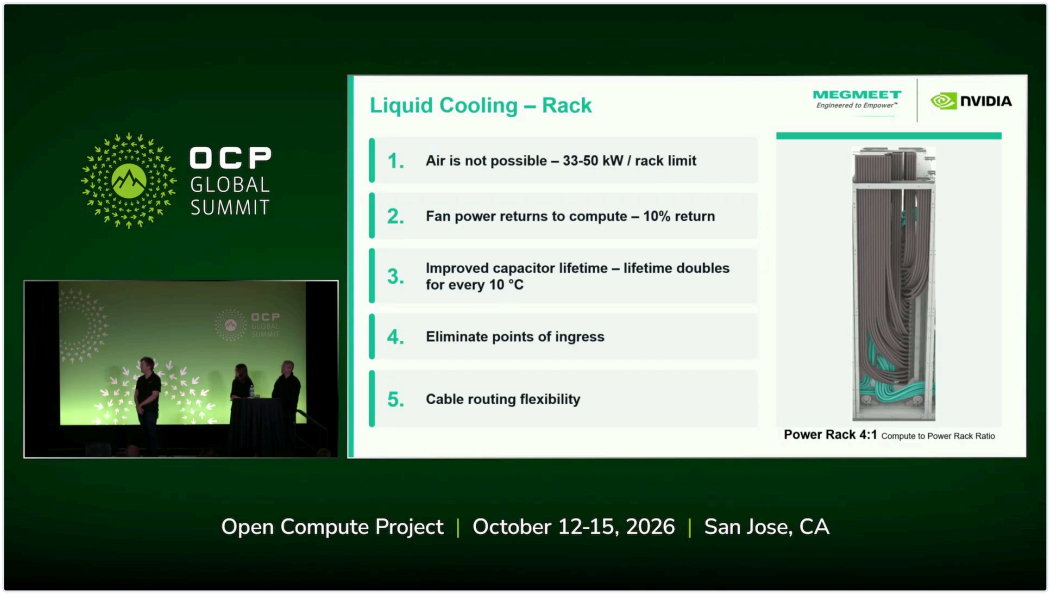
那么到第三代，我们为什么要转向液冷？转向液冷有几个原因。第一个是计算本身必须转向液冷。

> to 54-V DC on the 54-V bus bar. And initially, the first step, the first phase in that development is a power rack where we take in AC, and then we output 54-V DC. Or I'm sorry, we output 800-V, and then we take the 800-V and go to 54 V.

> And so what this allows us to do is long-term, anything that needs to be 800-V capable, whether it's support racks, networking racks, even our compute racks, right? As long as it's been designed around the MGX bus bar, we can take it and we can make it land in an AC data center or a 800 V DC data center. And so then for Gen3, why are we moving to liquid cooling? So there's several reasons why we're moving to liquid cooling. The first is the compute just has to move to liquid cooling.

> **小结**　把电源拆成独立机架，是这一页最关键的结构性决定：机架从"自带电源"变成"从母排取电"，于是交流与直流数据中心可以用同一套计算托盘。

05　液冷之于机架：五条理由



Liquid Cooling – Rack：五条理由——风冷已不可行（33–50 kW/机架上限）、风扇功耗还给计算（10%）、电容寿命每降 10 °C 翻倍、消除进尘点、线缆走线灵活；右侧为电源机架内部，标注 Compute to Power Rack Ratio 4:1

本质上，系统里需要的散热器那么多，根本没有足够的空间把所有东西都塞进去。第二个是，一旦你把整机架的功率算进来，机架里能散掉多少热，还有一个占地面积的限制。所以即便我们把电源机架开始用电源托盘填满，散热量仍然会撞上一个上限——大概 750 kW 到 1 MW。所以因为这个，我们必须转向液冷。

下一个原因是，那些被风扇白白消耗掉的功率，我们可以把它还给计算。另外具体到电源，我们一直用大量电解电容来做交流输入电流整形。把这些电容贴在冷板上，就能把它们的温度降下来——按照阿伦尼乌斯定律，电容每降温 10 °C，寿命就翻一倍。所以这是很划算的。

然后我们还能消除进尘点。凡是和数据中心电源打过交道的人都知道，一个常见失效模式就是灰尘导致短路，以及碎屑、施工杂物。如果我们能把 PSU 完全密封，这些失效模式就消除了。

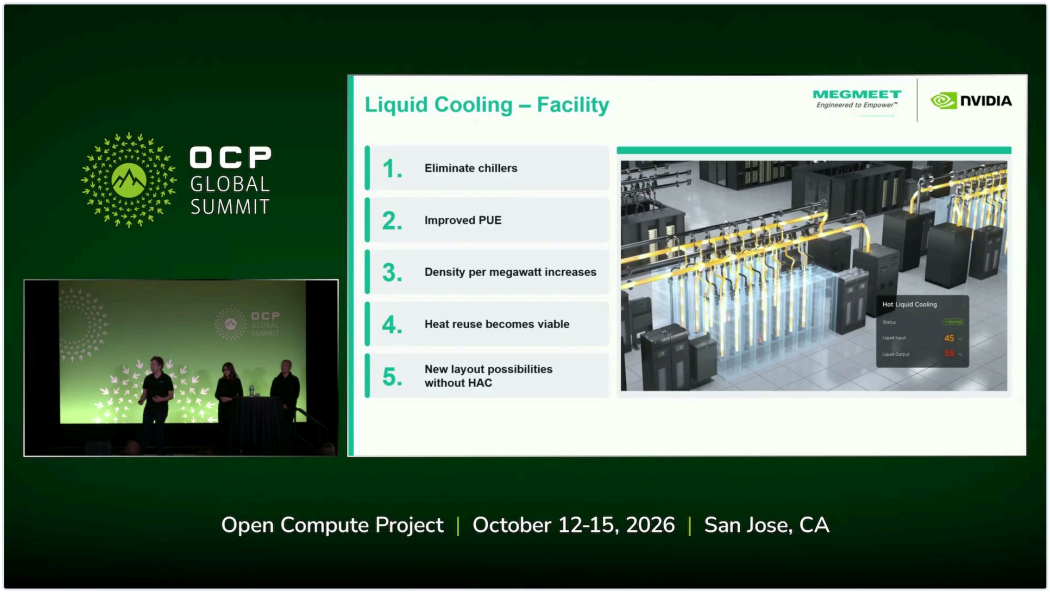
最后，我在这里展示的是一个电源机架的背面，完全插满电源托盘。你可以看到，要管理从所有这些电源线缆上方流出的热风，是完全做不到的——除非我们转向液冷电源托盘。

那么从机房的角度看，我们非常想提升整体制冷效率。做法是取消冷机。这就是为什么我们要走到 45 °C 的液体进液温度；通过改用干冷器闭式循环，我们可以……

> With all of the heat sinks that we need in the system, there's just not enough volume to fit everything. And the second thing is once you actually take the actual rack power, there's also just a footprint limitation on how much power you can dissipate in the rack. And so even if we were to for our power rack start filling it with power shelves, there's still going to be a limit of 750 kW to 1 MW of dissipation. And so then we need to move to liquid cooling for that reason. The next reason is all that fan power that we are dissipating, we can return that back to compute. And also with power supplies specifically, we've been using a lot of electrolytic capacitors for AC input current shaping. And so putting those against a cold plate allows us to take down their temperature which with the Arrhenius law, every 10 °C we can cool those capacitors, we double the lifetime. So that's beneficial. And then we also can eliminate points of ingress. So anyone who's worked around power supplies in data centers, a common failure mode is dust shorts and debris and work that's happening. And if we can completely seal the PSU, we eliminate those failure modes. And then finally what I'm showing here is the back of a power rack just completely filled with power shelves. And as you can see trying to manage the hot air flowing out over all those power cables, it's just completely unmanageable unless we move to liquid cooled power shelves. So then from a facility standpoint, we really want to improve the overall facility cooling efficiency. And so, we can do that by eliminating chillers. That's where we're moving to the 45° C liquid inlet temperature. And by moving to a dry

> **小结**　五条理由里只有第一条是"必须"，其余四条都是"顺便赚到"——液冷真正的门槛在于：不做，功率就上不去。

06　液冷之于机房：五条收益



Liquid Cooling – Facility：取消冷机、改善 PUE、每兆瓦密度提升、热回收可行、无热通道封闭的新布局；右侧机房照片带 45 °C 供水温度标注

……相比蒸发冷却，我们基本上每兆瓦每年可以省下 260 万加仑的水（口头表述）。所以这是一种有效得多的制冷方式，也让 PUE 更好。而且有了更好的制冷能力，我们还能提高每平方米的兆瓦数。热回收也变得可行。

我觉得比较有意思的一点是，我们实际上可以用新的方式改变数据中心的布局：不再需要操心热通道封闭。你现在可以把机架直接背靠背推在一起，甚至可以更容易地把机架放进集装箱里。

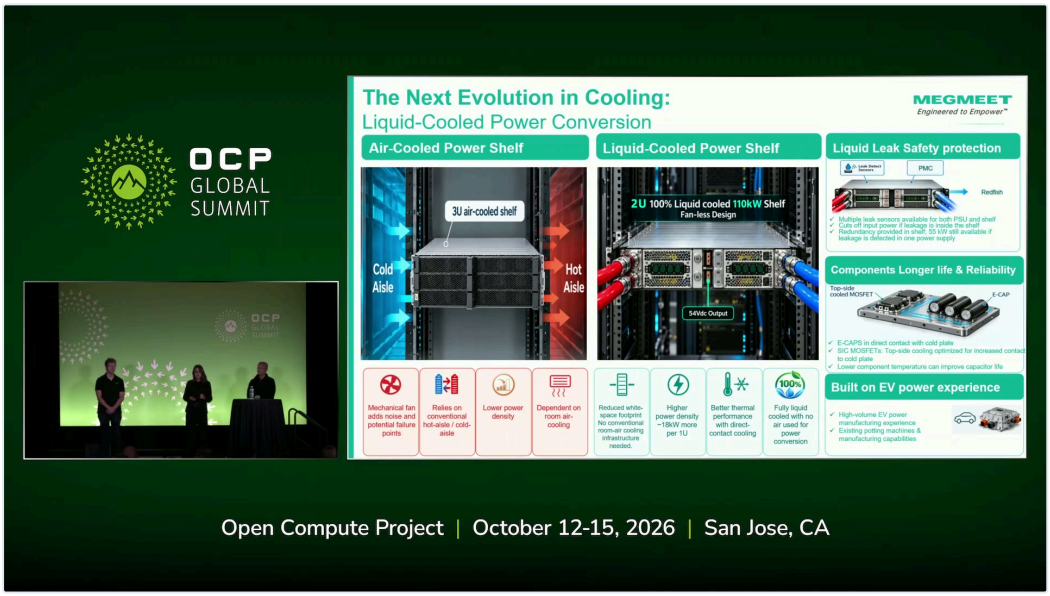
那么接下来，Roya 会多讲一些 MEGMEET 在做的工作。谢谢。

> chiller system with closed loops, we can basically save 2.6 million gallons per megawatt per year when compared to evaporative cooling, right? And so, it's a much more effective way to cool. That allows for a better PUE. And we can also then with the better cooling capability increase the megawatt per meter square. And heat reuse becomes a viable option. And I think one of the more interesting things is we can actually change the data center layout in new ways, right? We no longer have hot aisle containment that you need to worry about. And you can now push racks right back to back, or you can even put racks into shipping containers more easily. So, then up next Roya will talk more about the work that MEGMEET's doing. Thanks.

> **小结**　"机架可以背靠背、可以装进集装箱"听起来像小事，实际是把机房从"建筑设计"降级成"设备排布"——这是液冷对交付周期最实质的影响。

第三部分　电源托盘的液冷化

07　把电源托盘也做成液冷



The Next Evolution in Cooling: Liquid-Cooled Power Conversion：左为 3U 风冷电源托盘（风扇、冷热通道、四项缺点），右为 2U 100% 液冷 110kW 无风扇托盘（四项收益），最右三栏为漏液安全、器件寿命、EV 电源经验

我们都知道液冷进入计算托盘带来的好处。但随着机架密度越来越高，还有更多热方面的挑战需要解决。对我们来说，解决办法是把电源托盘也做成液冷。

我们这款 2U 110 kW 电源托盘是 100% 液冷的，内部零风量，是完全无风扇的设计。所以所有热量都通过机房的液冷回路带走。

这就从根本上改变了机架里、以及机房里的整套基础设施：不再有传统的冷热通道分区，不再依赖高速风扇和随之而来的噪音，这些全部被消除。

当我们在电源托盘里去掉风扇、风扇功耗以及风扇带来的所有失效点，我们实际上就能做出密度高得多的电源。同时机架空间、以及那里的白空间，也得到更有效的利用。

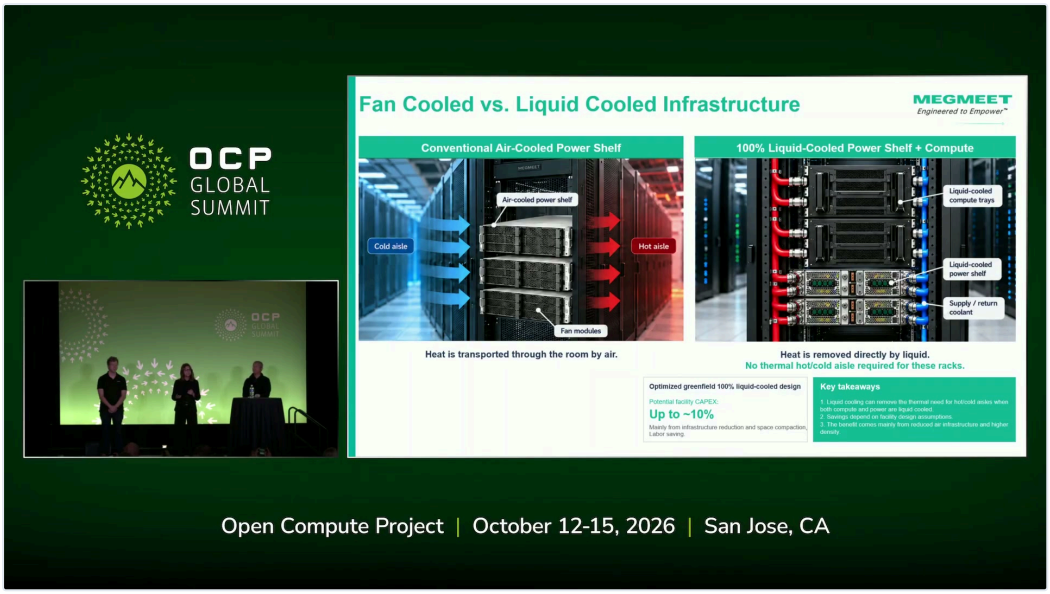
当把电子器件和液体放在一起，安全性上显然需要更稳健的做法。所以我们在电源和电源托盘两级都增加了非常多的漏液传感器。一旦电源托盘内部发生泄漏，它就会切断供电——要么是 55 kW 的单个电源，要么是 110 kW 的整个托盘。

我还想说，液冷还有更多好处。像电解电容和半导体这些器件，始终在更低的温度下运行，这提升了器件寿命，也提升了可靠性。最后，我们在电子制造、灌封设备等方面的经验，让我们能把这项技术更快推向市场。

> So, we all know about the benefits of liquid cooling coming into compute tray. But as we move forward to a much higher rack density, there is even more thermal challenges or concerns that needs to be addressed. For us, we solved this issue by making the power shelf liquid cooled. So, our 2U 110 kW power shelf is 100% liquid cooled with zero air flow inside. This is a complete fanless design. So, all that heat is extracted through the facility liquid cooling loop. And so, this fundamentally changes the whole infrastructure in the rack and in the data hall. So there is no more traditional hot and cold island data hall and relying on the high speed fans and all the noise that comes with it. Those are all eliminated. So coming into power shelf when we remove the fan and fan power and all the failure points that comes with the fan, we actually enable a much higher density power supply. And also more efficient use of the rack space and of course the white space that is there. So when we combine electronics and liquid there is obviously a much more robust approach to the safety. So we added so many more liquid sensors that is at power supply and power shelf level. So in case there is a leakage inside the power shelf, it shuts down the power either power supply which is 55 kilowatt or 110 kilowatt which is at power shelf level. I want to say that there is more benefits to the cooling. So all those components like e-cap and semiconductors are always running at lower temperature. So this increases the life of the components and also reliability. And finally our experience with e-manufacturing, potting machines and everything that we have today enables us to introduce this technology much faster to the market.

> **小结**　漏液检测被拆成两级：单电源泄漏只切单电源，托盘泄漏才切整个托盘——这是在用"可承受的失效粒度"换取液冷在机架里的准入资格。

08　风冷 vs 液冷：CAPEX 最多省 10%



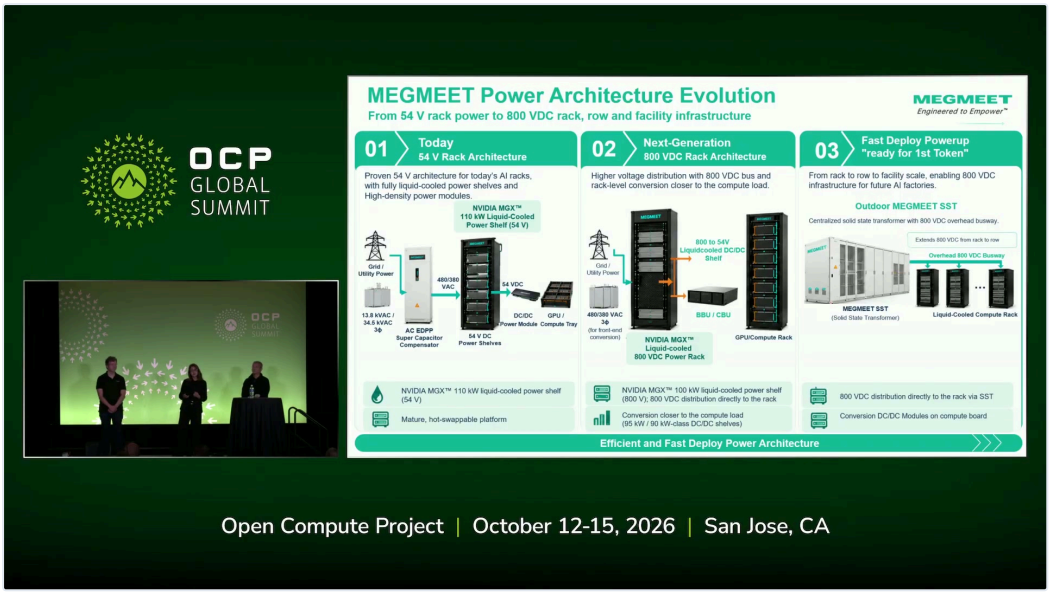
Fan Cooled vs. Liquid Cooled Infrastructure：左为常规风冷电源托盘（冷/热通道、风扇模块），右为 100% 液冷电源托盘 + 计算（供回液、无冷热通道）；下方为绿色方案 CAPEX 最多省约 10% 与三条结论

所以如果你把计算和电源托盘的功率变换器都改成液冷，我们的研究表明，CAPEX 最多可以省 10%。这主要来自白空间的节省、功率变换的紧凑化、人工的节省，以及那些随风冷而来的、现在不再需要的基础设施。

> So if you move both compute and power shelf power converters, both of them to liquid cooling, our study shows that you can actually save on capex up to 10% and that's mainly coming from white space saving, compact power conversion, and also labor saving, and all that infrastructure that comes with the air cooling that is not needed anymore.

> **小结**　10% 是这场演讲里唯一的经济性数字，而且它成立的前提很明确：计算和电源必须同时液冷，只改一边拿不到。

09　三级功率架构演进



MEGMEET Power Architecture Evolution：01 今天 54 V 机架架构（13.8/33/34.5 kVAC → AC SDPP → 480/380 VAC → 54 V DC 电源托盘 → PDB → GPU）；02 下一代 800 VDC 机架架构（800 转 54 V 液冷 DC/DC 托盘 + BBU/CBU）；03 快速部署 SST（室外固态变压器 → 架空 800 VDC 母排 → 液冷计算机架）

我们有一个成熟的 54 V 直流设计，已经为 NVIDIA MGX 系列发布了产品。同时我们也推出了纯电源机架，也就是 800 V 直流。我们的方案与计算托盘配套的所有 DC/DC 功率变换器和模块结合在一起。

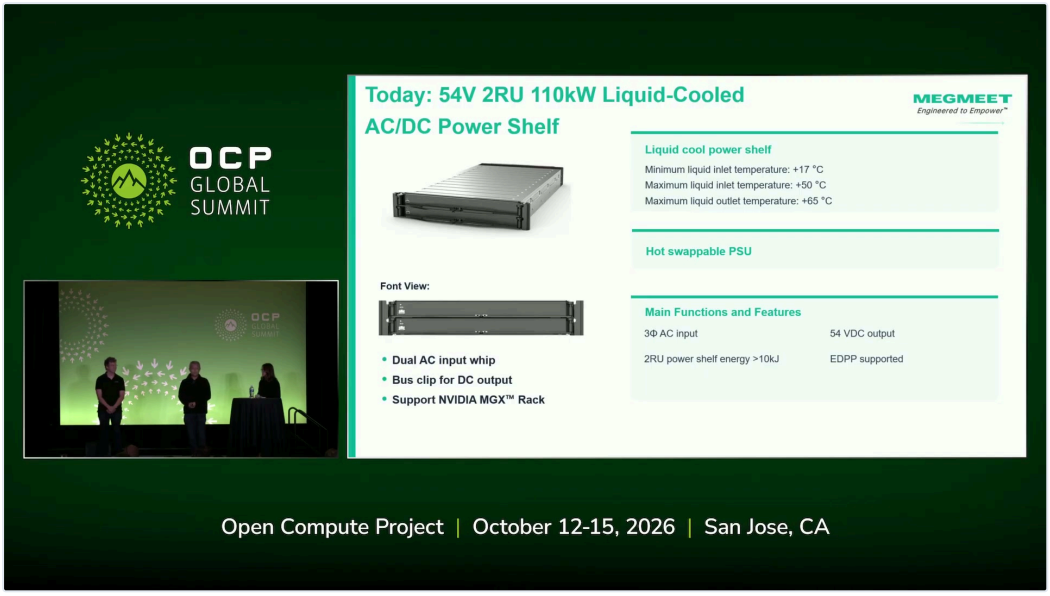
下一代，我们的做法是走到机房层级。也就是说，随着我们明年要发布的 SST（固态变压器），我们把整条 800 V 直流母排直接送到计算机架——计算机架同样是液冷的。这样我们就把任何下一代 AI 数据中心的部署变得快得多。

> So, we have a mature design that is at 54 V DC. We have released products for NVIDIA MGX series. And also introduce all the power only racks, which is 800 V DC. Our solution combines with all the DCDC power converters and modules that comes together with a compute tray. As a next generation, our approach is to go to facility level. So, that means with SST that we are going to release next year, we bring the whole 800 V DC bus bar directly to the compute rack, which is also liquid cooled. So, that's how we introduce and make it much faster to deploy for any next generation of AI data centers.

> **小结**　三步走的落点在第三步：SST 把变压器从室内搬到室外，800 V 母排变成架空走线——数据中心的供电边界因此从"房间"缩到"机架"。

第四部分　产品细节

10　今天的 54V 2RU 110kW 电源托盘



Today: 54V 2RU 110kW Liquid-Cooled AC/DC Power Shelf：进液 +17 °C / 最高 +50 °C、出液最高 +65 °C；热插拔 PSU；3Φ 交流输入、54 VDC 输出、>10 kJ 能量、EDPP 支持；双路交流输入线、直流母排夹、支持 NVIDIA MGX 机架

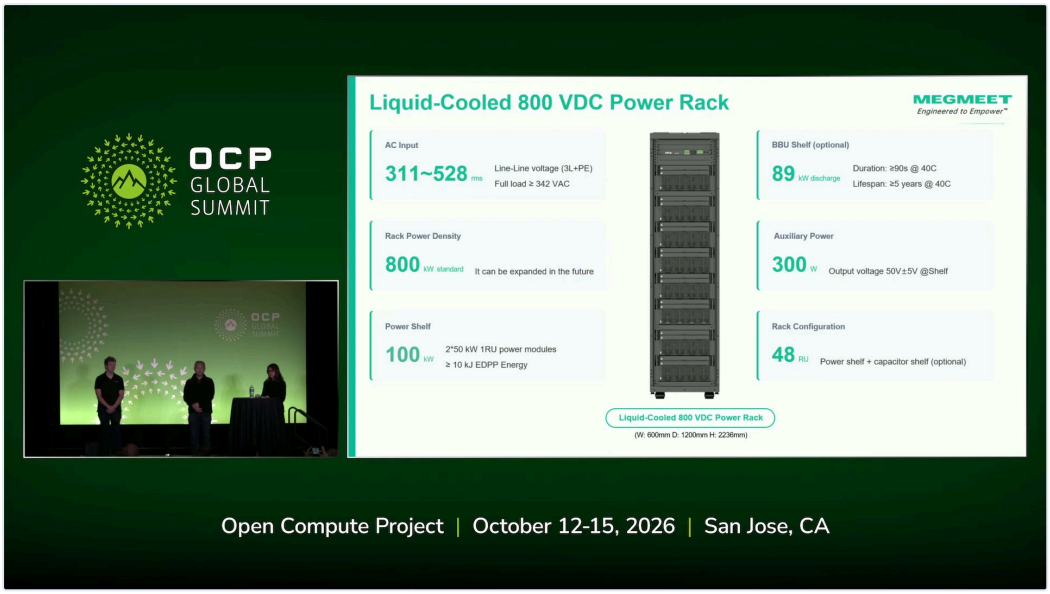
那么接下来，我把时间交给我的同事石磊。

好。我来介绍一些液冷方面的具体设计。第一个是 54 V、2RU、110 kW 的液冷电源托盘。这个托盘是 2RU，里面有两个 PSU，PSU 是热插拔的。托盘本身有双路交流输入，还有用于直流输出的母排夹（bus clip）。这个托盘可以支持 NVIDIA MGX 机架。它符合液冷规格，液体温度范围是 17 °C 到 50 °C。基本特性是：输入采用端子接入，同时它有 10 kJ 的内部能量，用于输入电流整形。

> With that, I'm passing to my colleague Shi Lei. Okay. I will introduce some specific partial design on the liquid cool. So, the first one is the 54 V 2RU 110 kW liquid cool shelf. So, this shelf is a 2RU, and we have two PSUs inside. So, the PSU actually is hot swappable. And also we have the shelf itself has dual AC inputs, and also it has bus clip for the DC output. And this shelf can support the NVIDIA MGX rack. So, it's for the liquid cool specs and the liquid temperature range is from 17° to 50°. The basic feature actually is the input is terminal input and also it has 10 K joules the internal energy for input current shaping.

> **小结**　注意进液温度上限给到 +50 °C，比机房侧常说的 45 °C 还高 5 度——这是为"用温水冷却"留出的余量，直接决定能不能取消冷机。

11　液冷 800 VDC 电源机架



Liquid-Cooled 800 VDC Power Rack：交流输入 311–528 Vrms（满载 ≥342 VAC）、机架功率密度 800 kW、单托盘 100 kW（2×50 kW 1RU 模块、≥10 kJ EDPP）、可选 BBU 89 kW 放电 / ≥90 s @40C / ≥5 年寿命、300 W 辅助电源、48 RU 配置、尺寸 600×1200×2236 mm

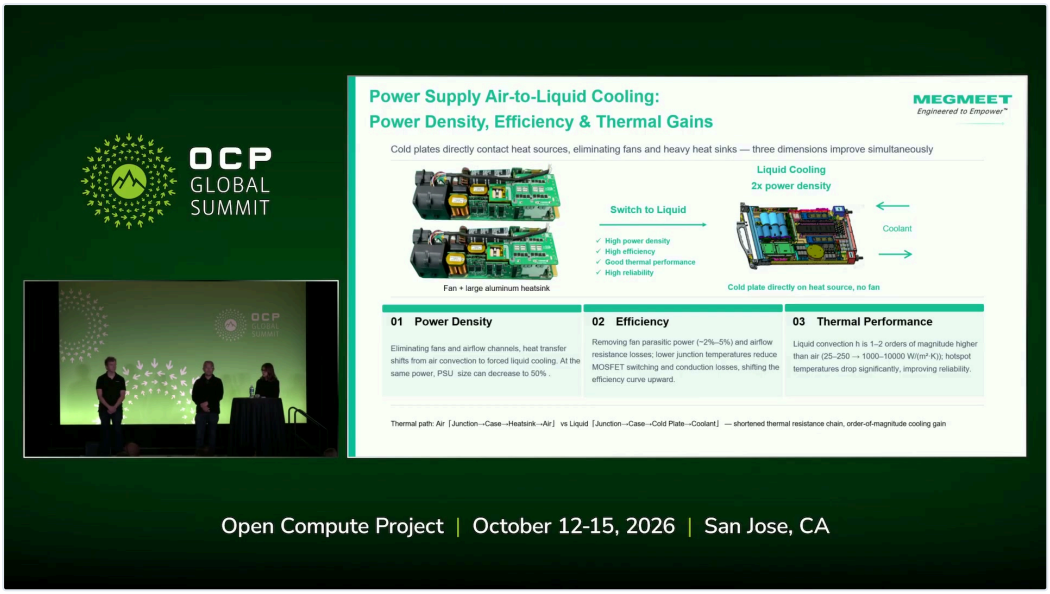
这是我们液冷的 800 V 直流产品。这个机架里实际有八个电源托盘，每个托盘可以输出 100 kW，所以总功率可以达到 800 kW。

我们还有可选的 BBU，可以用它提供 89 kW 的放电功率，同时把保持时间延长到 90 秒。这个产品还有 300 W 的辅助电源。我们还为这个机架提供电容树（capacitor tree），你也可以用电容树来做输入电流整形。

> Yeah, this is our liquid cool 800 V DC products. For this rack actually it has eight power shelves inside. So, each power shelf can deliver 100 kW power. So, the total power can reach to 800 kW. We have also the BBU select. You can use this BBU to deliver 89 kW discharge power and also it extends the hold up time to 90 seconds. This power shelf, this product also have 300 W auxiliary power. We also provide the capacitor tree for this rack so you can also use the capacitor tree to shaping the input current.

> **小结**　800 kW 机架配 90 秒 BBU，正好对上"电网需要时可以把负载推后"的调度逻辑——备电时长已经不是按断电时长设计，而是按切换窗口设计。

12　从风冷到液冷：三个维度同时改善



Power Supply Air-to-Liquid Cooling: Power Density, Efficiency & Thermal Gains：风冷（风扇 + 大铝散热器）与液冷（冷板直贴热源、无风扇、2 倍功率密度）对比；下分功率密度（同功率下体积可降至 50%）、效率（省 2%–5% 风扇寄生功耗）、热性能（液体对流换热系数高 1–2 个数量级）三栏

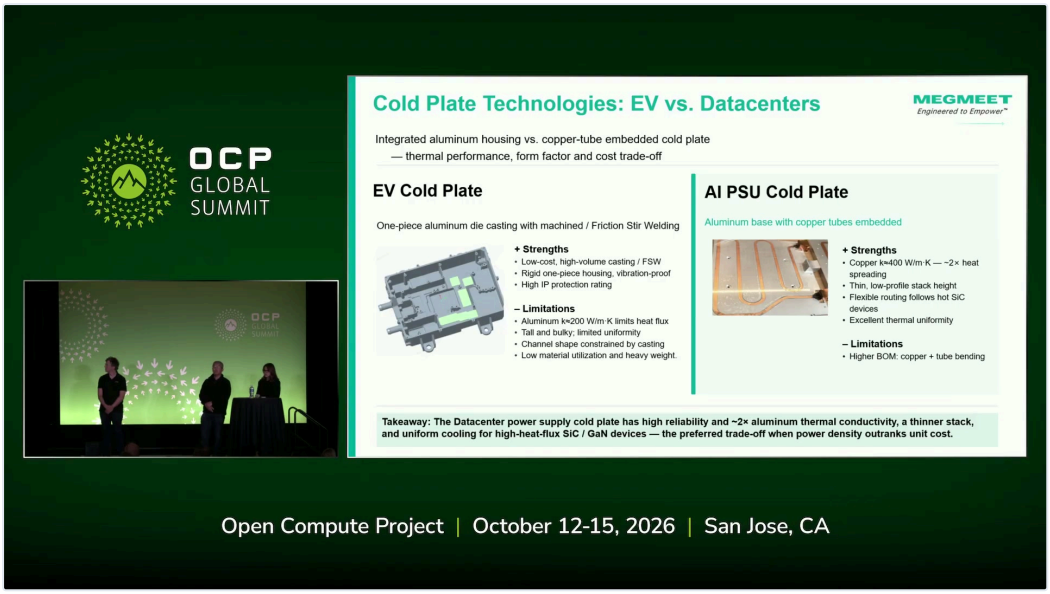
这一页是展示从风冷到液冷的升级。左边是风冷电源，它有风扇，还有笨重的铝散热器。右边是液冷电源的一个例子。在液冷电源里，我们去掉了风扇，也去掉了风道。所有功率器件都贴在 PSU 底部的冷板上。用这种配置，它可以做到很高的功率密度，尺寸也能更小。有了这个好处，我们可以很轻松地放更多电容，在内部获得更高的储能。

由于这种配置去掉了风扇，我们省下了风扇功耗；而且功率器件——比如 MOSFET——的结温可以降低，所以整体效率可以更高。另外你也知道，液体的换热系数高得多。所以用液冷，热点可以被压住，从而实现更低的温度和更高的可靠性。

> This page just to show the upgrade from the air cooling to liquid cooling. You see on the left side is the air cool power supply. So, it has a fan and it also has bulky aluminum heat sinks. On the right side is just an example of the liquid cool power supply. So, in the liquid cool power supply, we remove the fan. And also remove the air channel. All the power components attached to the cold plate at the bottom of the PSU. So, with this configuration, it can reach high power density and can get a smaller size. So, with this benefit, we can easily put more capacitors and get a higher energy storage inside. Since this configuration removes the fan, we save fan power and also the power components actually can decrease the junction temperature such as the MOSFETs. So, the total efficiency can be higher. And also you know that the liquid heat transfer coefficient is much higher. So, with this liquid cooling the hot spot can be suppressed. So, it can realize a lower temperature and also high reliability.

> **小结**　三个维度不是并列关系：省掉风扇腾出的体积才装得下更多电容，多出的电容才换来 10 kJ 的保持能量——体积红利被换成了电网侧的韧性。

13　冷板工艺：电动车 vs 数据中心



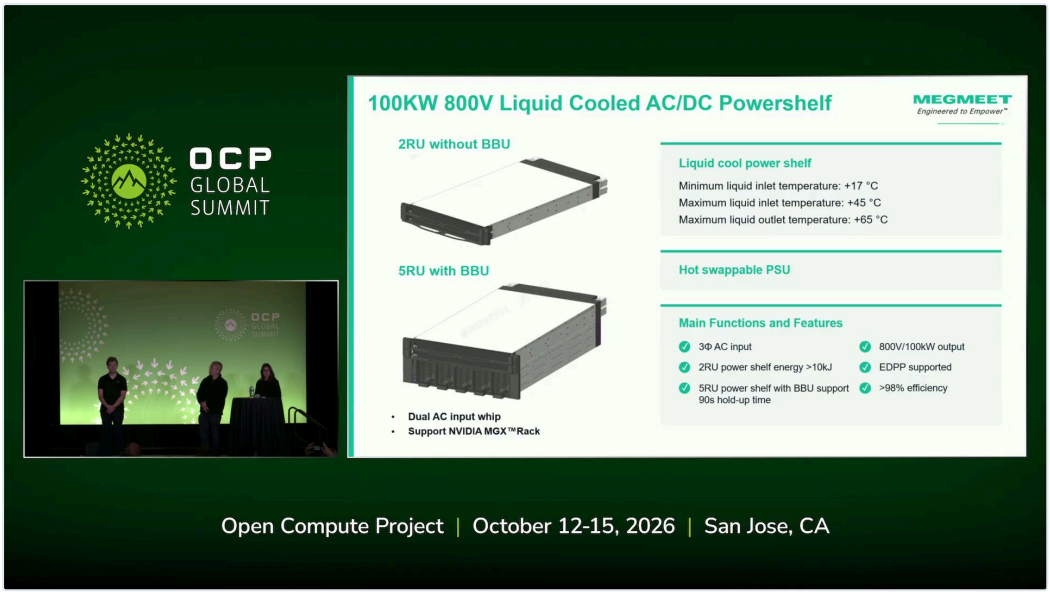
Cold Plate Technologies: EV vs. Datacenters：左侧电动车冷板（一体压铸 + 机加工 / 搅拌摩擦焊，κ≈200 W/mK，厚重大件、流道受铸造限制）；右侧 AI 电源冷板（铝基体嵌铜管，κ≈400 W/mK、更薄、走线可随 SiC 器件弯曲）；底部为取舍结论

这一页是拿电动车的冷板和电源的冷板做对比。电动车的冷板其实又笨重又厚。所以我们现在用的是嵌入铜管的冷板。这种冷板可以做得更薄，而且铜管的散热效果好得多——怎么说呢——所以我们可以用它把温度降得更低。

> This just to compare between the EV cold plate and the AI PSU cold plate. So, the EV cold plate actually is bulky and heavy. And so we now use the cold plate with embedded tubes. So, this cold plate actually we can make it thin and also the copper tube — how to say this — has much better heat dissipation, so we can use that one to get lower temperature.

> **小结**　电动车冷板追求"便宜、耐振、大批量"，AI 电源冷板追求"薄、导热、贴合热点"——同一项工艺，在两边的优化目标几乎相反。

14　100kW 800V 液冷 AC/DC 电源托盘



100KW 800V Liquid Cooled AC/DC Powershelf：2RU 无 BBU 与 5RU 带 BBU 两种形态；进液 +17 °C、最高 +45 °C、出液最高 +65 °C；3Φ 交流输入、800V/100kW 输出、>10 kJ、EDPP、5RU 支持 90 秒保持、效率 >98%

这是我们为 100 kW、800 V 液冷 AC/DC 电源托盘在做的东西。我们有两个版本：一个是 2RU 带 BBU 的版本，另一个是 5RU 带 BBU 的。2RU 不带 BBU 的版本其实也有两个 PSU 在里面，PSU 本身是热插拔的。5RU 只是在 2RU 托盘的基础上增加了 BBU 模块。

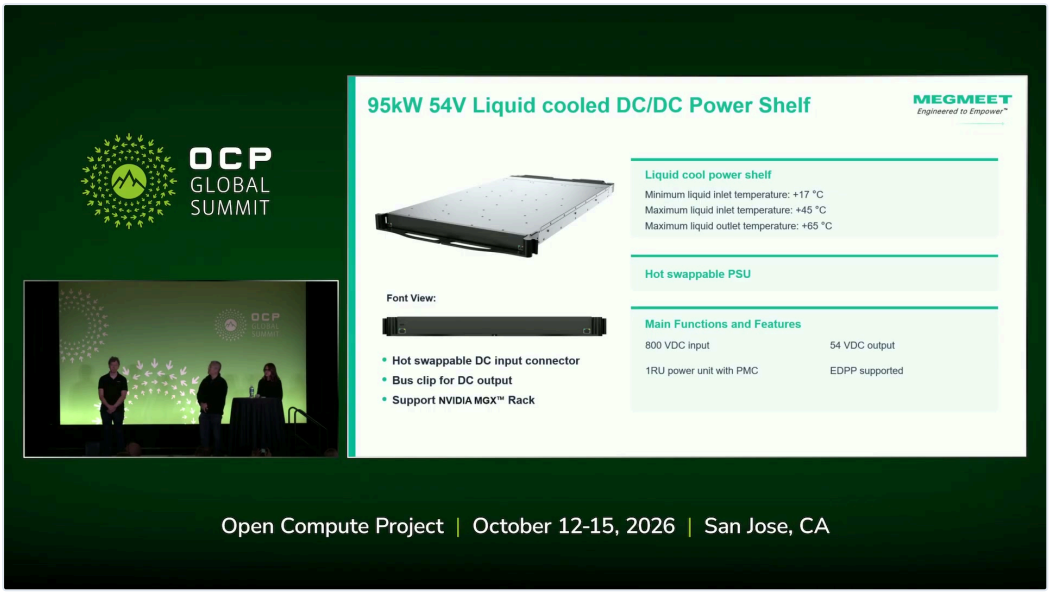
托盘本身有双路交流输入，也能支持 NVIDIA MGX 机架。液体进液温度范围是 17 °C 到 45 °C。2RU 电源托盘的内部能量是 10 kJ。

好，下一个。这是另一个产品，95……

> This is what we are doing for the 100 kW 800 V liquid cool AC/DC power shelf. So, we have two versions. One is 2RU with BBU version and another one is the 5RU with BBU. And the 2RU without BBU actually is also — there's two PSU inside, so the PSU itself is hot swappable. The 5RU just — there is just the additional BBU modules to this 2RU shelf. So the shelf itself have dual AC input and it can also support a NVIDIA MGX rack. The liquid inlet temperature range is 17 °C to 45 °C. The 2RU power shelf on the internal energy is 10 kJ. Okay. Next one. This is another product that is 95

> **小结**　注意进液温度从 54 V 那款的 +50 °C 收紧到 +45 °C：电压升高后器件应力更大，温水冷却的余量被让掉了 5 度。

15　95kW 54V 液冷 DC/DC 电源托盘



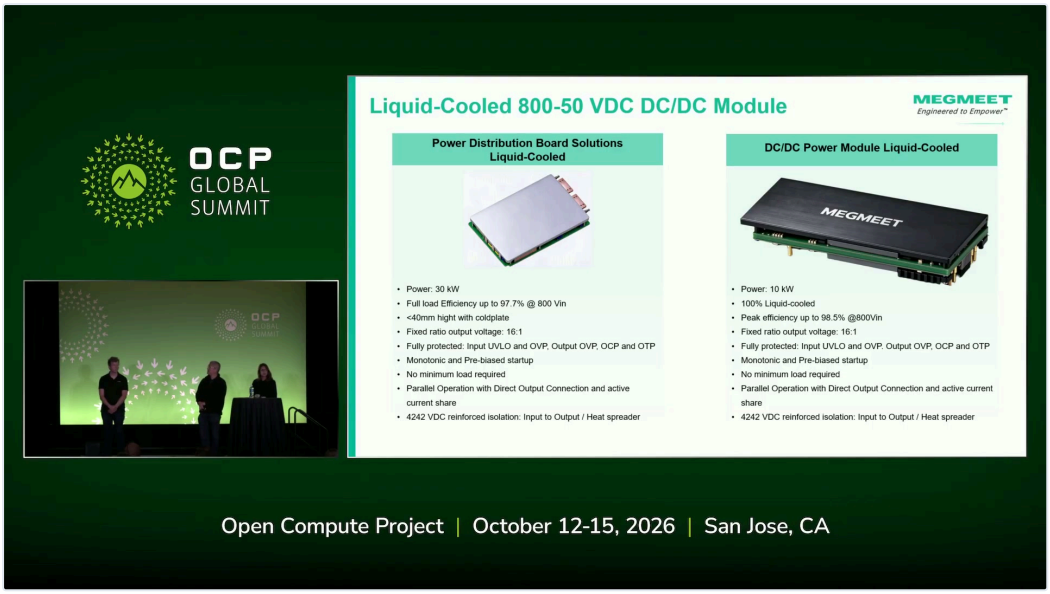
95kW 54V Liquid cooled DC/DC Power Shelf：1RU 单 PSU、热插拔、800 VDC 输入 / 54 VDC 输出、带 PMC、EDPP 支持；热插拔直流输入连接器、直流输出母排夹、支持 NVIDIA MGX 机架

……kW、54 V 的液冷 DC/DC 电源托盘。它其实是配合 800 V 电源托盘使用的。这个托盘实际上就是一个 PSU，所以它也是热插拔的。它同样支持 NVIDIA MGX 机架。输入是 800 V，输出是 54 V。

> kW 54 V liquid cool DCDC power shelf. Actually, this one is used with the 800 V power shelf. So, this shelf actually is one PSU. So, this one is also hot swappable. It can also support the Nvidia MGX rack. So, the input is 800 V. So, output is 54 V.

> **小结**　1RU 做到 95 kW，意味着 800 V 直流进来、54 V 出去的这一级，已经被压到和传统 PSU 差不多的体积里。

16　800–50 VDC 液冷 DC/DC 模块



Liquid-Cooled 800-50 VDC DC/DC Module：左侧 PDB 液冷方案（30 kW、满载效率 97.7% @800Vin、<40mm 高度、16:1 固定变比）；右侧 DC/DC 液冷模块（10 kW、100% 液冷、峰值效率 98.5% @800Vin、16:1）

这是 DC/DC 模块，800 V 到 50 V。左边第一个是 PDB 模块，它是液冷的，总功率 30 kW。右边第二个是 10 kW 的 DC/DC 模块。

好，如果大家有任何问题，或者对我们的产品感兴趣，可以到 C19 展位，我们可以详细聊。谢谢。

> This is the DC-DC modules, it's 800 V to 50 V. So, the first one, the left one is the PDB module and it's liquid cooling. The total power is 30 kW. And the second one, the right one is the 10 kW DC-DC module. Yeah, if you have any question or if you are interested in our products, you can go to C19 booth and we can have a detailed discussion. Thank you.

> **小结**　16:1 固定变比、98% 级的效率、模块化并联——这一级已经彻底"服务器化"，电源不再是基础设施，而是板级部件。

附录

A. 关键数据核对表

数据项	幻灯片值	口播值	本文采用
Gen 1（GB200/GB300 NVL72）	145 kW，54 VDC	145 kW	一致

数据项	幻灯片值	口播值	本文采用
Gen 2（Vera Rubin NVL72）	330 kW，54 VDC	330 kW	一致
Gen 3	570 kW，54 VDC，800 VDC 进 PSU	570 kW	一致
母排电压	54 VDC	"50-V bus bar"	**幻灯片值**（口播有误）
风冷机架功率上限	33–50 kW / rack	—	幻灯片值
电源机架散热上限	—	750 kW – 1 MW	口播值（已标注）
风扇功耗可回收比例	10%	"还给计算"	幻灯片值
电容寿命	每降 10 °C 翻倍	同	一致
干冷器节水	—	260 万加仑 / MW / 年	口播值（已标注）
机房进液温度	45 °C（示意图）	45 °C	一致
液冷电源托盘	2U，110 kW，100% 液冷，零风量	2U 110 kW	一致
单 PSU 功率	—	55 kW	口播值（已标注）
CAPEX 节省	最多约 10%	最多 10%	一致
54 V 托盘（今日）	2RU 110 kW，进液 +17~+50 °C，出液 ≤+65 °C，>10 kJ	17–50 °C	一致
800 VDC 电源机架	800 kW，8×100 kW，48 RU	8 托盘 ×100 kW = 800 kW	一致
机架交流输入	311–528 Vrms，满载 ≥342 VAC	—	幻灯片值
BBU 放电	89 kW，≥90 s @40C，≥5 年	89 kW / 90 s	一致
辅助电源	300 W，50 V±5 V	300 W	一致
冷板体积缩减	同功率下可降至 50%	"尺寸能更小"	幻灯片值
风扇寄生功耗	约 2%–5%	—	幻灯片值
换热系数	25–250 → 1000–10000 W/(m²K)	"高得多"	幻灯片值

数据项	幻灯片值	口播值	本文采用
冷板导热率	铝 ≈200、铜 ≈400 W/mK	"铜好得多"	幻灯片值
100 kW 800 V 托盘	2RU（无 BBU）/ 5RU（带 BBU），进液 +17~+45 °C，效率 >98%	17–45 °C	一致
95 kW 54 V DC/DC 托盘	1RU，800 VDC 入 / 54 VDC 出	800 V → 54 V	一致
PDB 模块	30 kW，97.7% @800Vin，<40 mm	30 kW	一致
DC/DC 模块	10 kW，98.5% @800Vin	10 kW	一致
变比 / 隔离	16:1，4242 VDC 加强隔离	—	幻灯片值

B. 时间轴

时间	内容
00:00	开场与三位讲者
00:45	800 VDC 的必要性：功耗趋势与铜域扩张
03:03	四代电源平台与电源机架方案
03:57	为什么转向液冷（机架层面五条理由）
05:48	机房层面的五条收益
07:00	MEGMEET：把电源托盘做成液冷
09:21	风冷 vs 液冷：CAPEX 约 10%
09:51	三级功率架构演进
10:42	54 V 2RU 110 kW 液冷托盘
12:12	液冷 800 VDC 电源机架
13:15	风冷到液冷的三个维度
15:03	冷板工艺对比
15:42	100 kW 800 V 液冷 AC/DC 托盘
17:01	95 kW 54 V 液冷 DC/DC 托盘

时间	内容
17:42	800–50 VDC 液冷 DC/DC 模块

C. 术语对照

缩写	全称	说明
SST	Solid State Transformer	固态变压器，把中压交流直接变成 800 V 直流，可置于室外
PSU	Power Supply Unit	单个电源模块，可热插拔
Power Shelf	—	电源托盘，多个 PSU 组成的机架单元
PDB	Power Distribution Board	计算托盘内的配电板
VRM	Voltage Regulator Module	电压调节模块，最终供给 GPU 核心电压
BBU / CBU	Battery / Capacitor Backup Unit	电池 / 电容备电单元，用于掉电保持
EDPP	Energy During Power-loss Protection	掉电保持能量指标，单位 kJ
PMC	Power Management Controller	电源管理控制器
MGX	Modular GPU Accelerator	NVIDIA 机架级模块化架构规范，已贡献给 OCP
HAC	Hot Aisle Containment	热通道封闭
PUE	Power Usage Effectiveness	数据中心能源使用效率